Chapter 4. APPLICATIONS OF DERIVATIVES

(Page 341-485, Calculus Volume 1)



MAE101

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APPLICATIONS OF DIFFERENTIATION

4.1 Related Rates

In this section, we will learn:

How to compute the rate of change of one quantity in terms of that of another quantity.

RELATED RATES

Example 1

Air is being pumped into a spherical balloon so that its volume increases at a rate of $100\ cm^3/s$.

How fast is the radius of the balloon increasing when the diameter is 50 cm?

Example 1

The key thing to remember is that rates of change are derivatives.

- In this problem, the volume and the radius are both functions of the time t.
- The rate of increase of the volume with respect to time is the derivative dV / dt.
- The rate of increase of the radius is dr / dt.

RELATED RATES

Example 1

To connect *dV*/*dt* and *dr*/*dt*, first we relate *V* and *r* by the formula for the volume of a sphere:

$$V = \frac{4}{3}\pi r^3$$

Example 1

To use the given information, we differentiate each side of the equation with respect to *t*.

To differentiate the right side, we need to use the Chain Rule:

$$\frac{dV}{dt} = \frac{dV}{dr}\frac{dr}{dt} = 4\pi r^2 \frac{dr}{dt}$$

RELATED RATES

Example 1

Now, we solve for the unknown quantity:

$$\frac{dr}{dt} = \frac{1}{4\pi^2}\frac{dV}{dt}$$

If we put $r = 25$ and $dV/dt = 100$ in this equation, we obtain:

$$\frac{dr}{dt} = \frac{1}{4\pi(25)^2}100 = \frac{1}{25\pi}$$

The radius of the balloon is increasing at the rate of $1/(25\pi) \approx 0.0127$ cm/s.

Example 2

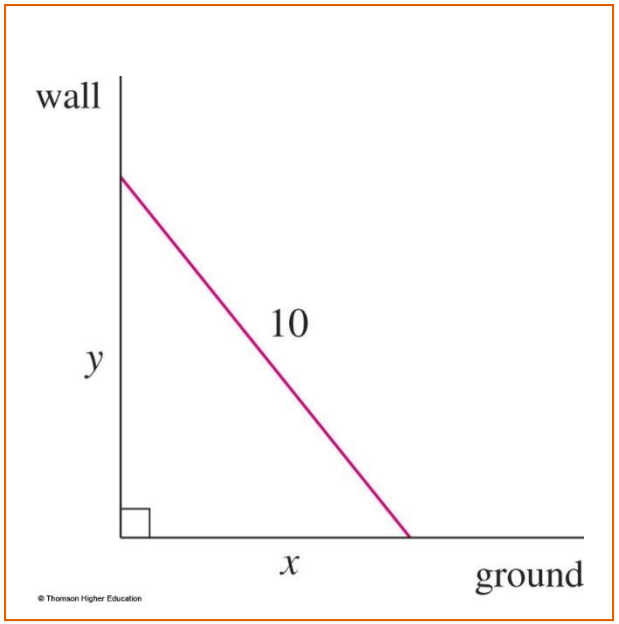
A ladder 10 ft long rests against a vertical wall. If the bottom of the ladder slides away from the wall at a rate of 1 ft/s, how fast is the top of the ladder sliding down the wall when the bottom of the ladder is 6 ft from the wall?

Example 2

We first draw a diagram and label it as in the figure.

- Let x feet be the distance from the bottom of the ladder to the wall and y feet the distance from the top of the ladder to the ground.

- Note that x and y are both functions of t (time, measured in seconds).



Example 2

We are given that

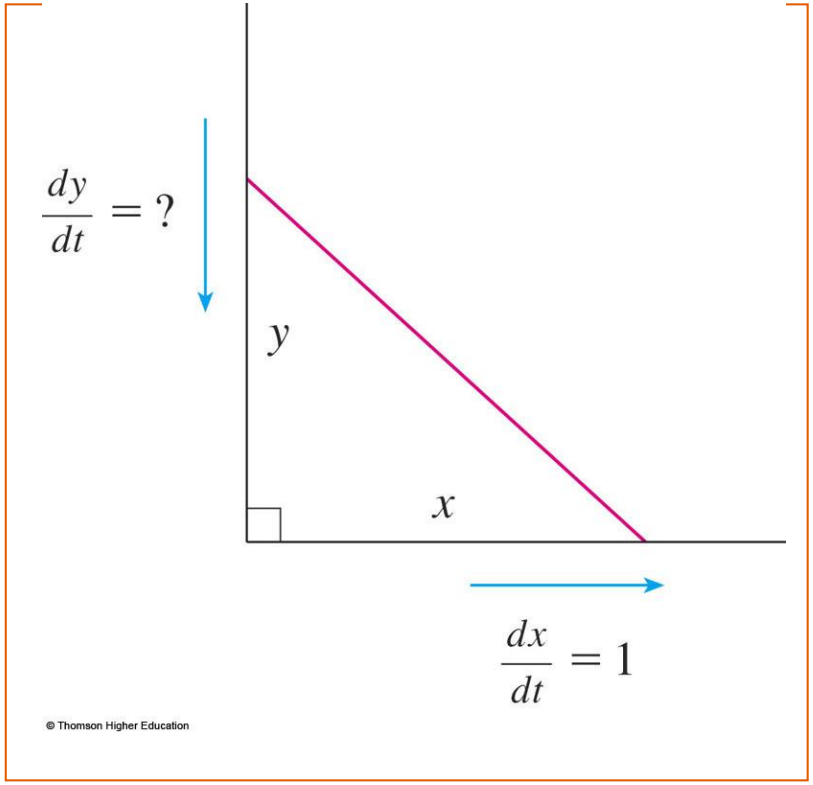
$dx / dt = 1$ ft/s

and we are asked

to find dy / dt

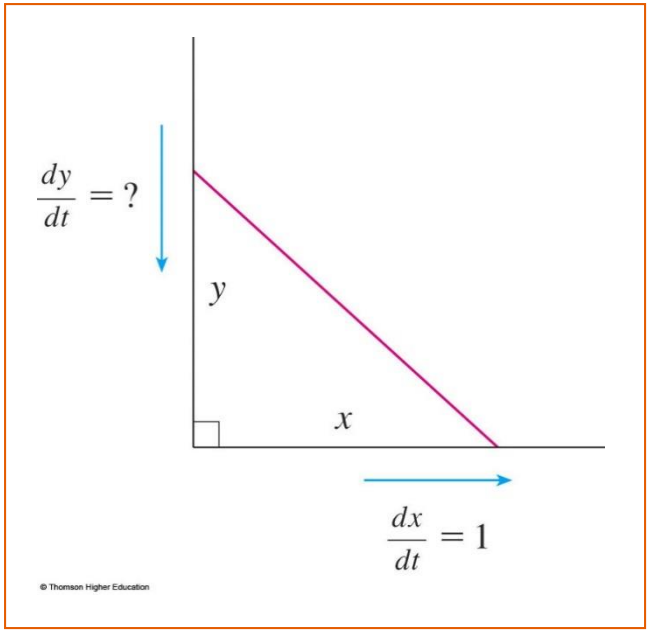
when

$x = 6$ ft.



Example 2

In this problem, the relationship between x and y is given by the Pythagorean Theorem: $x^2 + y^2 = 100$



Example 2

Differentiating each side with respect to *t* using the Chain Rule, we have:

$$2x\frac{dx}{dt}+2y\frac{dy}{dt}=0$$

– Solving this equation for the desired rate, we obtain:

$$\frac{dy}{dt}=-\frac{x}{y}\frac{dx}{dt}$$

Example 2

When $x = 6$, the Pythagorean Theorem gives $y = 8$ and so, substituting these values and
$dx / dt = 1$, we have:

$$\frac{dy}{dt} = -\frac{6}{8}(1) = -\frac{3}{4}\ ft / s$$

- The fact that dy / dt is negative means that the distance from the top of the ladder to the ground is decreasing at a rate of ¾ ft/s.
- That is, the top of the ladder is sliding down the wall at a rate of ¾ ft/s.

APPLICATIONS OF DIFFERENTIATION

4.2 Linear Approximations and Differentials

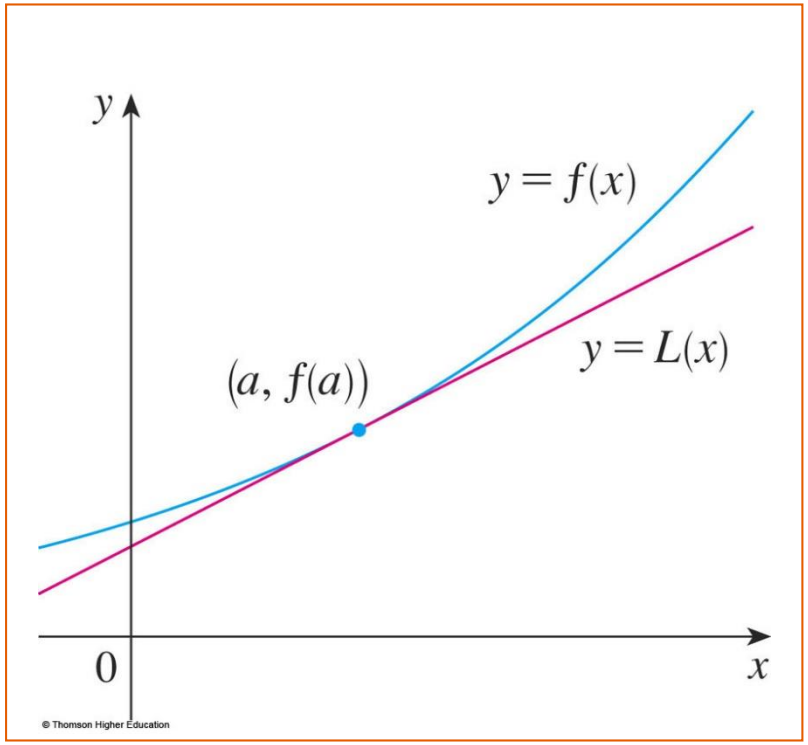
In this section, we will learn about:
Linear approximations and differentials and their applications.

LINEAR APPROXIMATIONS

We use the tangent line at $(a, f(a))$ as an approximation to the curve $y = f(x)$ when x is near a.

An equation of this tangent line is

$$L(x) = y = f(a) + f'(a)(x - a)$$



LINEAR APPROXIMATION

The approximation

$$f(x) \approx f(a) + f'(a)(x - a) = L(x)$$

is called the linear approximation of f at a.

LINEAR APPROXIMATIONS

Example 1

Find the linearization of the function $f(x) = \sqrt{x+3}$ at $a = 1$ and use it to approximate the numbers $\sqrt{3.98}$ and $\sqrt{4.05}$

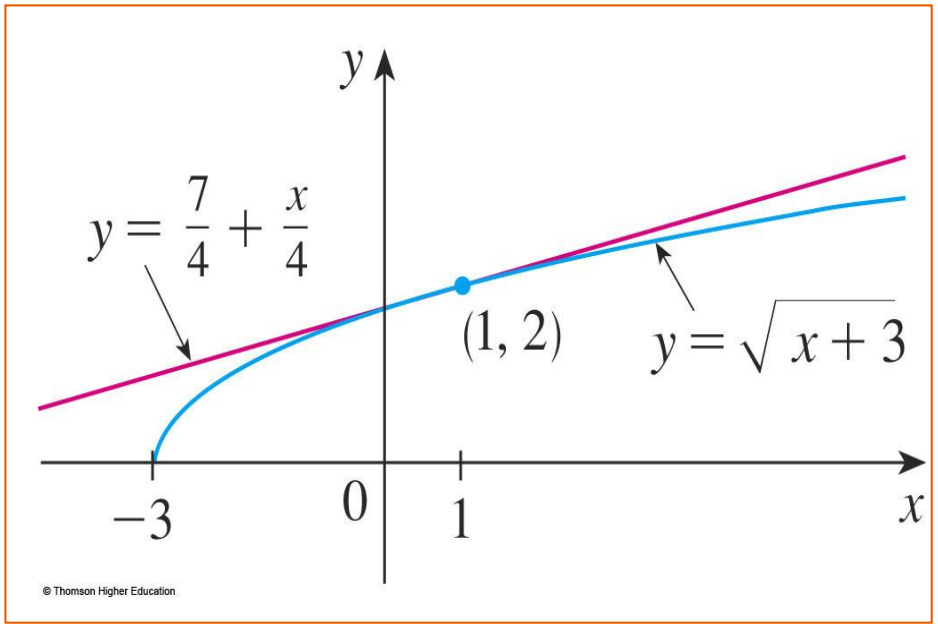
Are these approximations overestimates or underestimates?

LINEAR APPROXIMATIONS

Example 1

Putting these values into Equation 2, we see that the linearization is:

$$L(x) = f(1) + f'(1)(x-1)$$
$$= 2 + \tfrac{1}{4}(x-1)$$
$$= \frac{7}{4} + \frac{x}{4}$$



LINEAR APPROXIMATIONS

Example 1

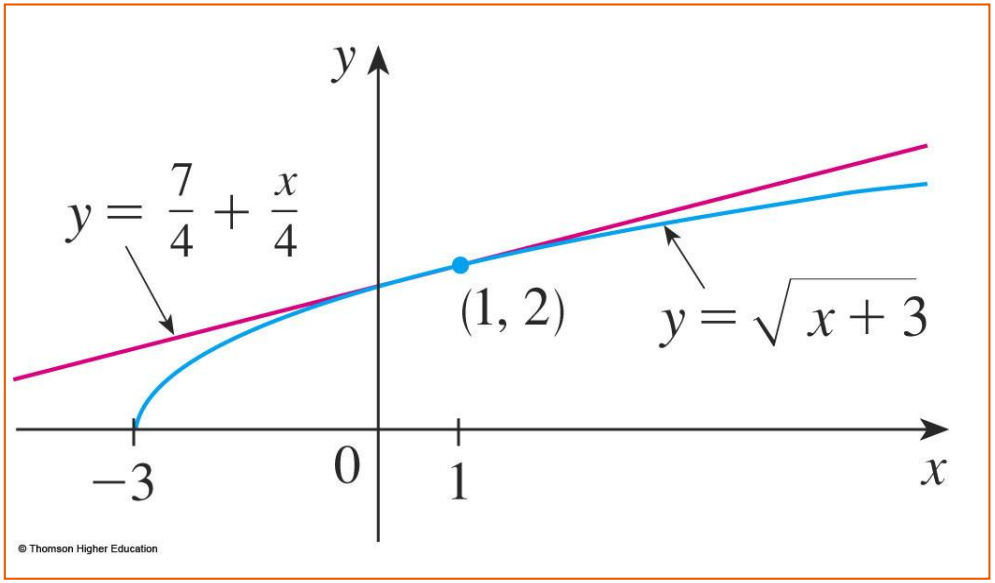
The corresponding linear approximation is: $\sqrt{x+3} \approx \frac{7}{4} + \frac{x}{4}$

(when x is near 1)

In particular, we have:

$$\sqrt{3.98} \approx \frac{7}{4} + \frac{0.98}{4} = 1.995$$

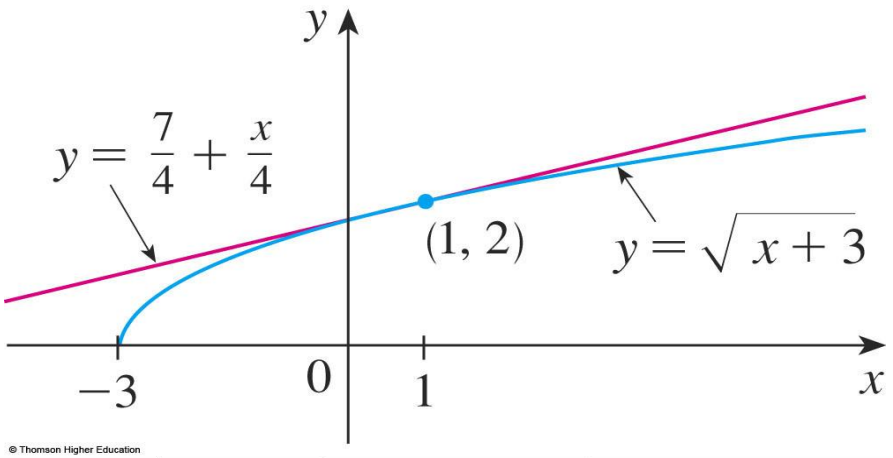
$$\sqrt{4.05} \approx \frac{7}{4} + \frac{1.05}{4} = 2.0125$$



LINEAR APPROXIMATIONS

Look at the table and the figure.

- The tangent line approximation gives good estimates if x is close to 1.
- However, the accuracy decreases when x is farther away from 1.



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	x	From $L(x)$	Actual value
$\sqrt{3.9}$	0.9	1.975	1.97484176 . . .
$\sqrt{3.98}$	0.98	1.995	1.99499373 . . .
$\sqrt{4}$	1	2	2.00000000 . . .
$\sqrt{4.05}$	1.05	2.0125	2.01246117 . . .
$\sqrt{4.1}$	1.1	2.025	2.02484567 . . .
$\sqrt{5}$	2	2.25	2.23606797 . . .
$\sqrt{6}$	3	2.5	2.44948974 . . .

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Use the linear approximation to of the function $f(x)=\sqrt{x+1}$ at a=3 to approximate the numbers. Select the correct ones.

a) $\sqrt{3.95}\approx 1.9875$

b) $\sqrt{4.05}\approx 2.0125$

c) $\sqrt{3.95}\approx 1.98746$

d) $\sqrt{4.05}\approx 2.01246$

DIFFERENTIALS

Example 2

The radius of a sphere was measured and found to be 21 cm with a possible error in measurement of at most 0.05 cm.

What is the maximum error in using this value of the radius to compute the volume of the sphere?

DIFFERENTIALS

Example 2

If the radius of the sphere is r, then its volume is $V = 4/3\pi r^3$.

- If the error in the measured value of r is denoted by $dr = \Delta r$, then the corresponding error in the calculated value of V is ΔV.

DIFFERENTIALS

Example 2

This can be approximated by the differential

$$dV = 4\pi r^2 dr$$

When $r = 21$ and $dr = 0.05$, this becomes:

$$dV = 4\pi(21)^2\, 0.05 \approx 277$$

The maximum error in the calculated volume is about 277 cm^3.

RELATIVE ERROR

Note

Relative error is computed by dividing the error by the total volume:

$$\frac{\Delta V}{V} \approx \frac{dV}{V} = \frac{4\pi r^2 dr}{\frac{4}{3}\pi r^3} = 3\frac{dr}{r}$$

Thus, the relative error in the volume is about three times the relative error in the radius.

RELATIVE ERROR

Note

In the example, the relative error in the radius is approximately $dr/r = 0.05/21 \approx 0.0024$ and it produces a relative error of about 0.007 in the volume.

The errors could also be expressed as percentage errors of 0.24% in the radius and 0.7% in the volume.

Quiz

Find the limit if $g(x) = \ln x$

$$\lim_{x \to 2} \frac{g(x) - g(2)}{x - 2}$$

a. 1 b. ½ c. 1/3 d. 2

Quiz

Find $\frac{dy}{dt}$ when x=2 if $\frac{dx}{dt}=1,$ $y=x^3-3x^2$

a. 1 b.2 c.3 d.4 e.0

Quiz

The equation of the tangent line to the curve y= x - sin x at (∏, ∏) is

a) y=0 b) y=2x c) y=2x-2

d) y=2x+2 e) None of the others

Quizzes

If $1/f = 1/u + 1/v$ then du/dv is:

a) $f/(v-f)^2$

b) $-f^2/(v-f)^2$

c) $2f^2/(v-f)^2$

d) $-2f^2/(v-f)^2$

e) None of the others

APPLICATIONS OF DIFFERENTIATION

4.3 Maxima and Minima

In this section, we will learn:

How to find the maximum

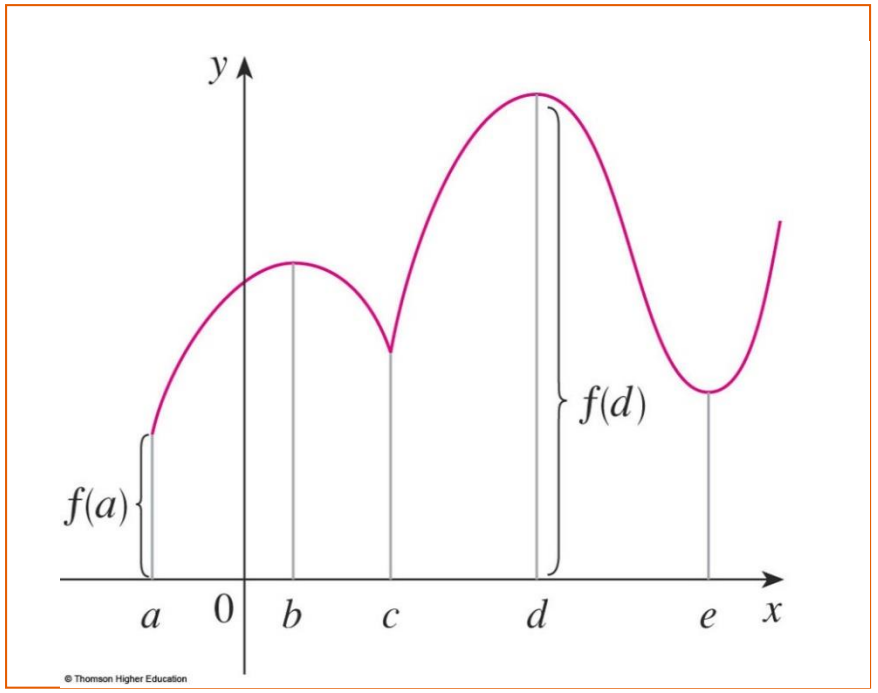
and minimum values of a function.

MAXIMUM & MINIMUM VALUES

Definition 1

A function f has an **absolute maximum** (or global maximum) at c if $\boldsymbol{f(c) \geq f(x)}$ **for all** $\boldsymbol{x}$ **in** $\boldsymbol{D}$, where D is the domain of f.

The number $f(c)$ is called the **maximum value of** $\boldsymbol{f}$ **on** $\boldsymbol{D}$.

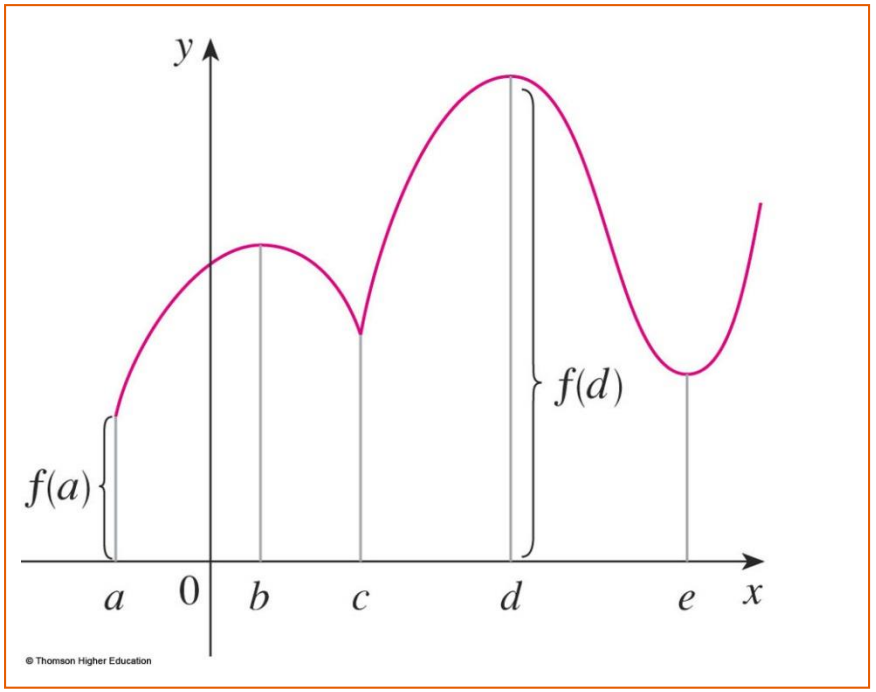


MAXIMUM & MINIMUM VALUES

Definition 1

Similarly, f has an **absolute minimum** at c if $f(c) \le f(x)$ for all x in D and the number $f(c)$ is called the minimum value of f on D.

The maximum and minimum values of f are called the **extreme values** of f.

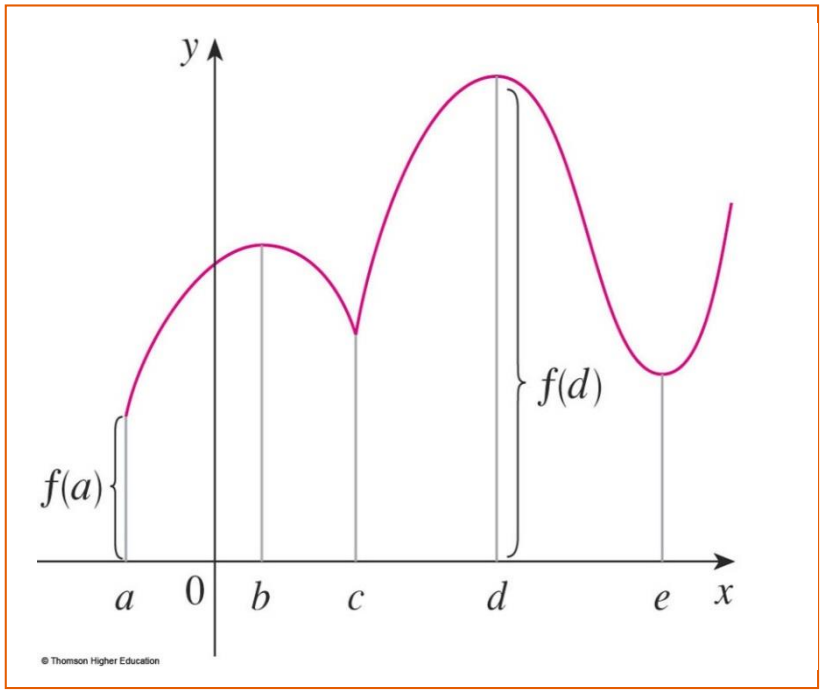


MAXIMUM & MINIMUM VALUES

Definition 2

– A function *f* has a **local maximum** (or relative maximum) at *c* if $f(c) \geq f(x)$ when ***x* is near *c***.

– Similarly, *f* has a **local minimum** at *c* if $f(c) \leq f(x)$ when *x* is near *c*.



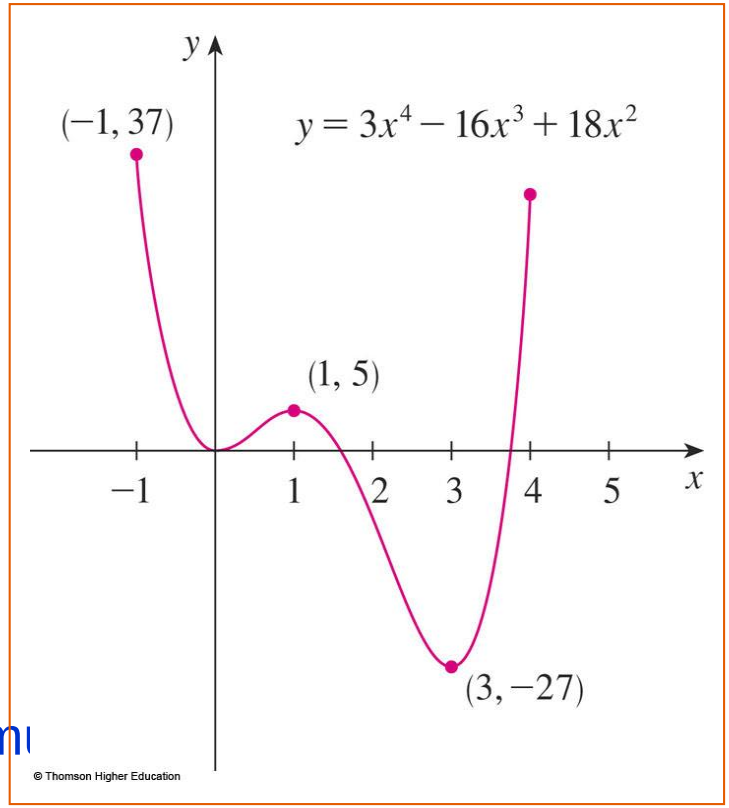
MAXIMUM & MINIMUM VALUES

Example

The graph of the function

$$f(x) = 3x^4 - 16x^3 + 18x^2, \quad -1 \le x \le 4$$

is shown here.



$f(1) = 5$ is a local maximum

the absolute maximum is $f(-1) = 37$

$f(0) = 0$ is a local minimum and

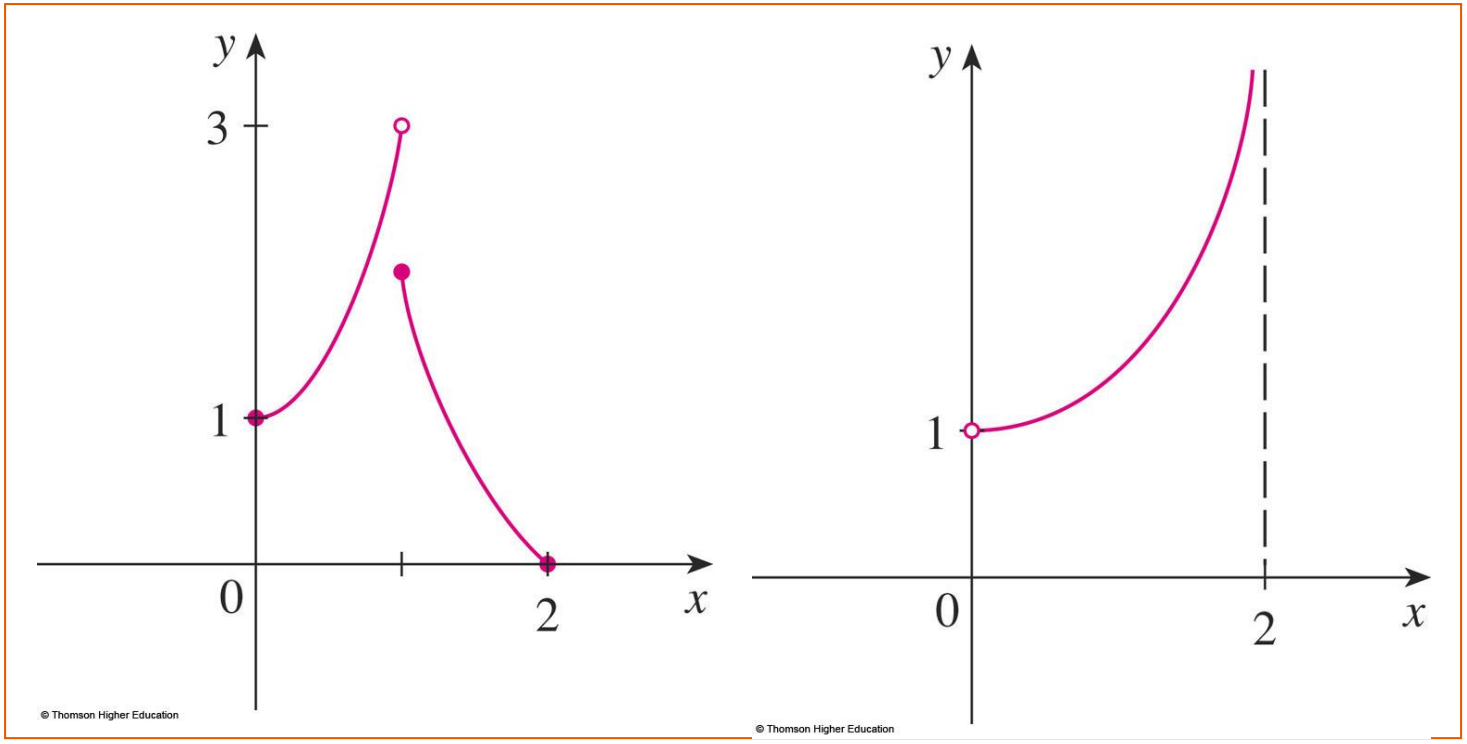
$f(3) = -27$ is both a local

and an absolute minim

EXTREME VALUE THEOREM

In the first figure, why isn't 3 the absolute maximum value?

In the second, does it have the absolute maximum and minimum value?

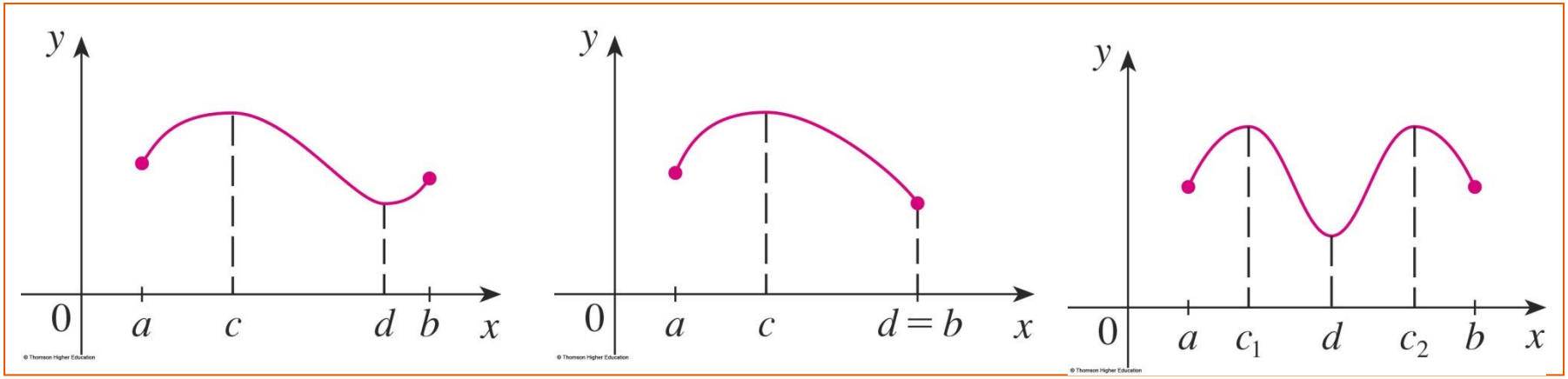


MAXIMUM & MINIMUM VALUES

Theorem

If f is continuous on a closed interval [a, b], then f attains an absolute maximum value f(c) and an absolute minimum value f(d)

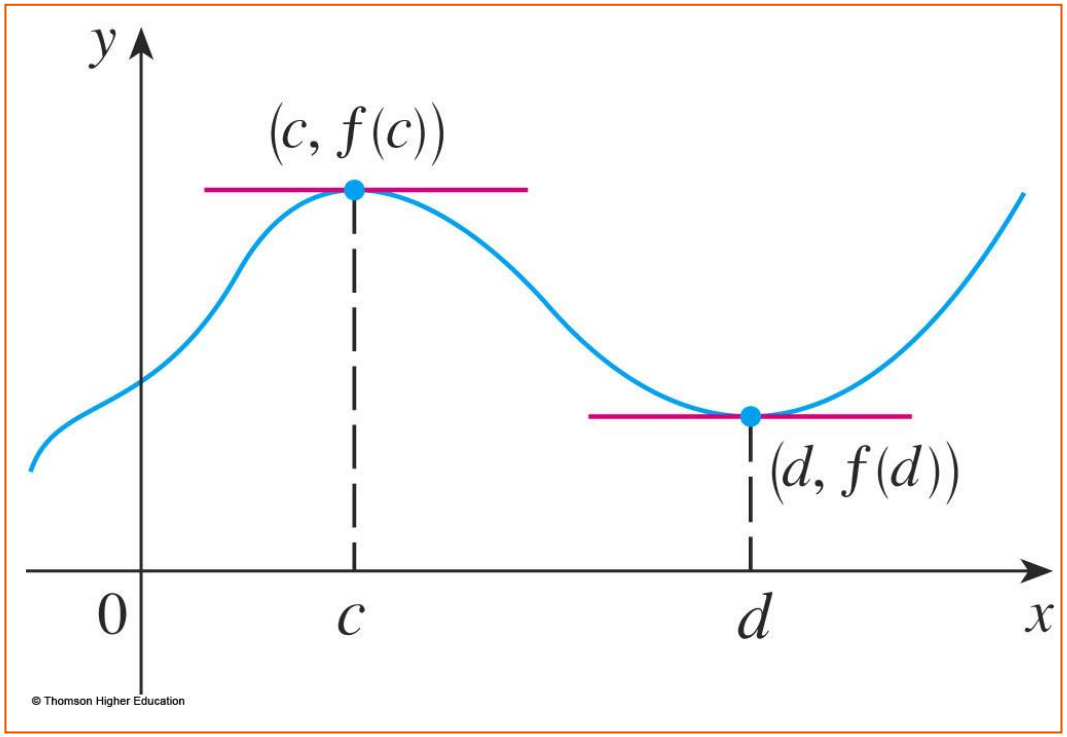
at some numbers c and d in [a, b].



MAXIMUM & MINIMUM VALUES

The theorem does not tell us how to find these extreme values.

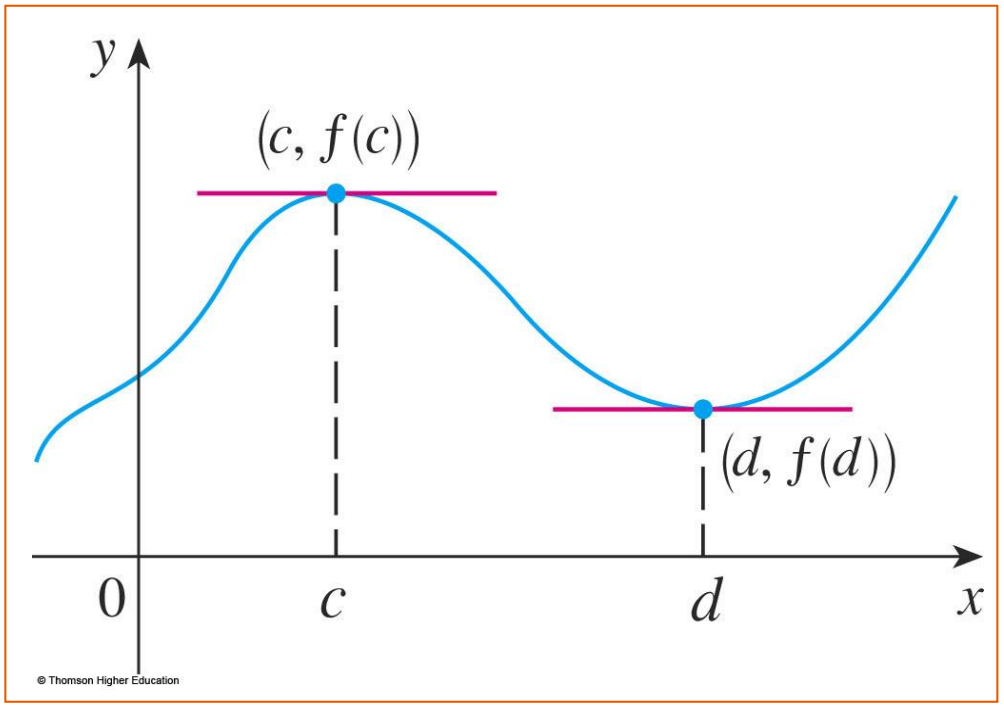
– We start by looking for local extreme values.



FERMAT'S THEOREM

Theorem

If f has a local maximum or minimum at c, and if f ′(c) exists, then f ′(c) = 0.



Is it true if say that

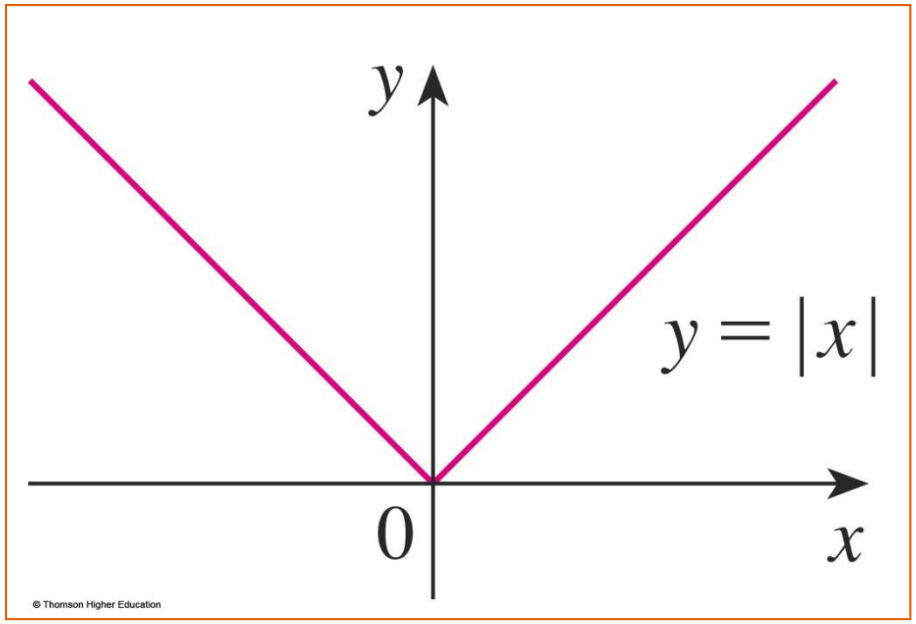
” f’ (c)=0 if f has local extreme value at c?”

CRITICAL NUMBERS

Example

– The function $f(x) = |x|$ has its (local and absolute) minimum value at 0.

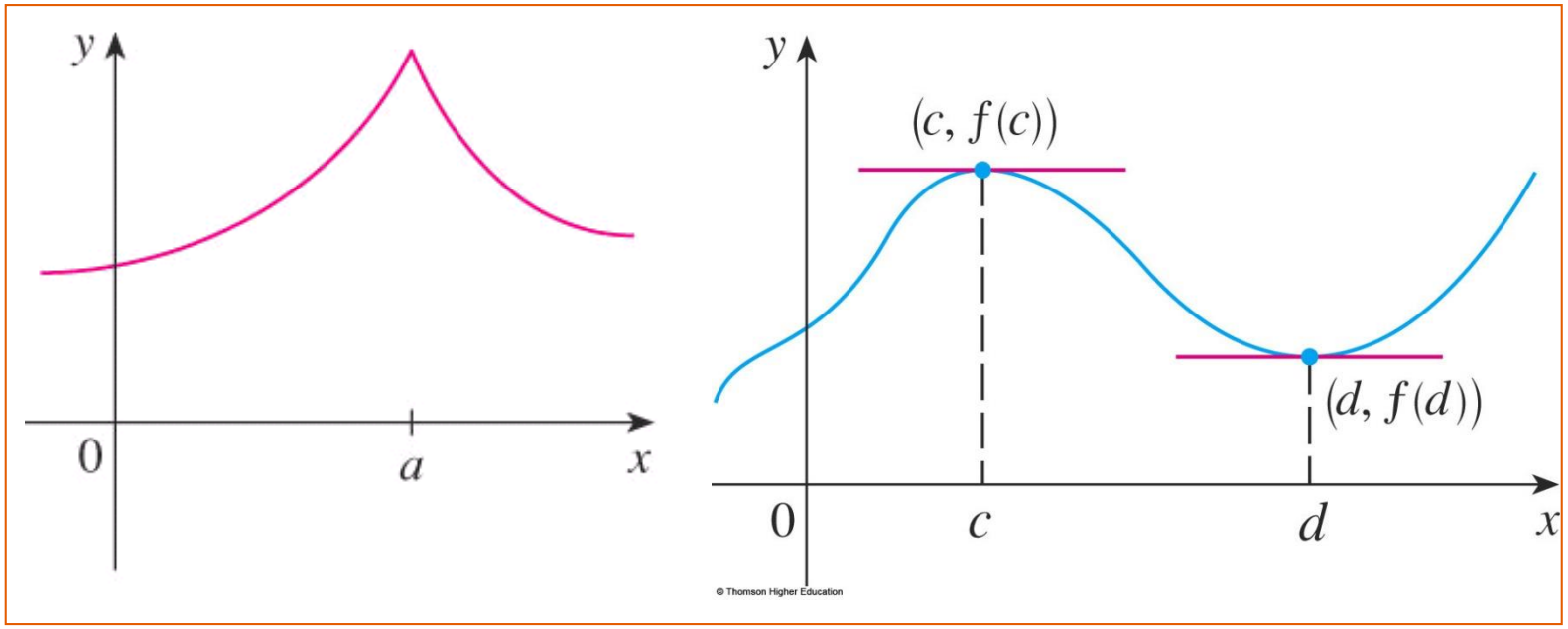
– $f'(0)$ does not exist.



CRITICAL NUMBERS

Definition

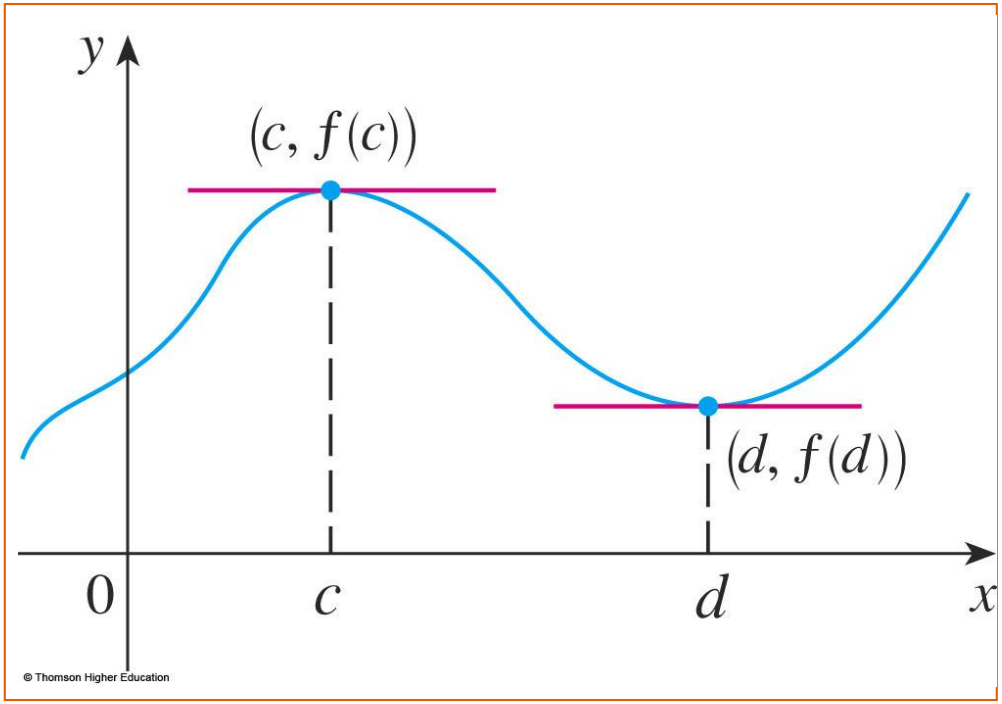
A **critical number (giá trị tới hạn)** of a function *f* is a number *c* in the domain of *f* such that either $f'(c) = 0$ or $f'(c)$ does not exist.



CRITICAL NUMBERS

Theorem

If f has a local maximum or minimum at c, then c is a critical number of f.



CLOSED INTERVAL METHOD

To find the **absolute** maximum and minimum values of a continuous function f on a closed interval $[a, b]$:

1. Find the values of f at the critical numbers of f in (a, b).
2. Find the values of f at the endpoints of the interval.
3. The largest value from 1 and 2 is the absolute maximum value. The smallest is the absolute minimum value.

Select the correct ones.

a. If f’ (c)=0 then f has the local maximum or minimum at c.

b. If f has the absolute minimum value at c then f’ (c)=0.

c. If f is continuous on (a,b) then f attains an absolute maximum value f(c) and an absolute minimum value f(d) for some c and d in (a,b).

d. All of the above.

e. None of the above.

APPLICATIONS OF DIFFERENTIATION

4.4 The Mean Value Theorem

In this section, we will learn about:

The significance of the mean value theorem.

ROLLE'S THEOREM

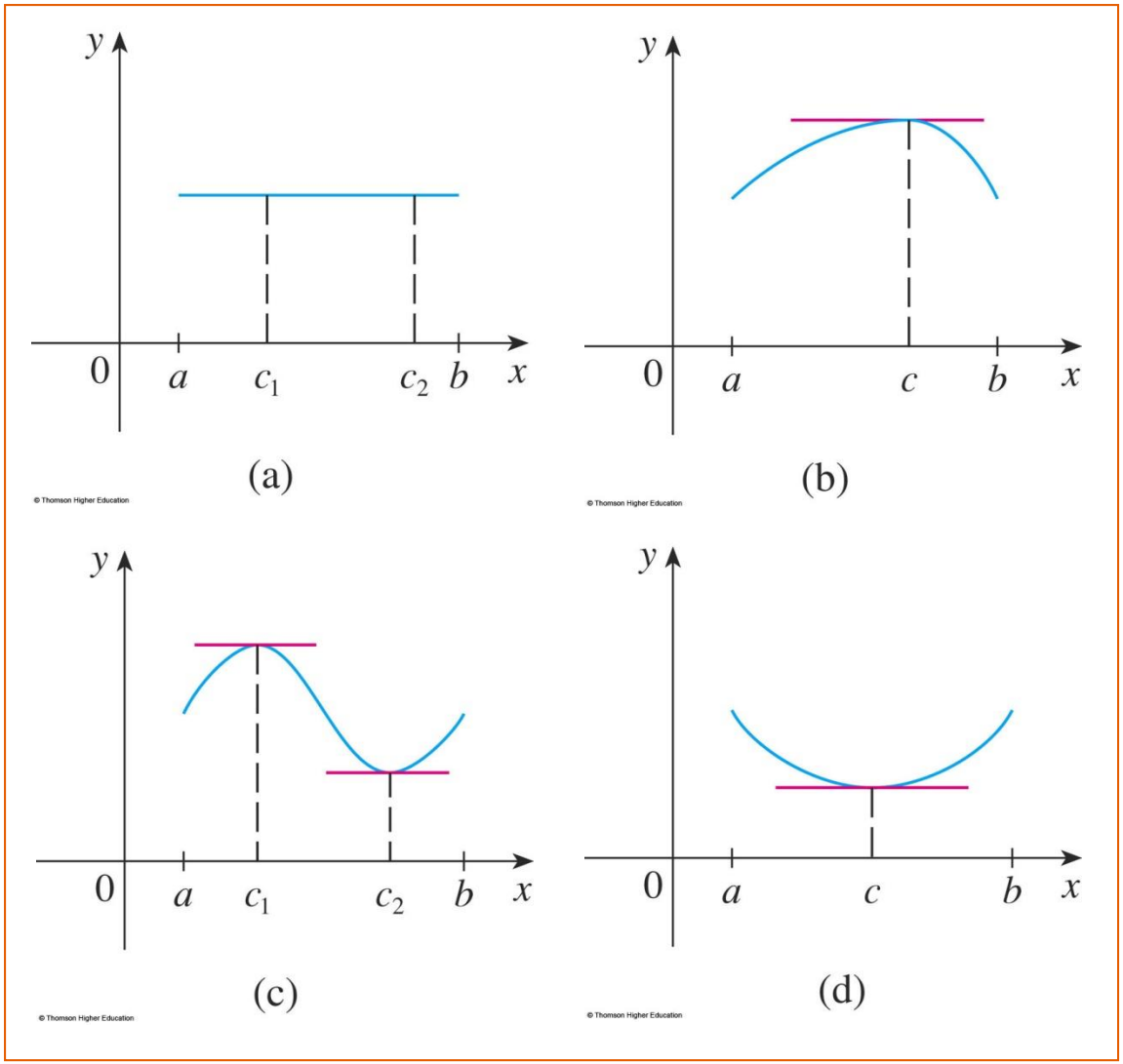
Let f be a function that satisfies the following three hypotheses:

1. f is continuous on the closed interval $[a, b]$
2. f is differentiable on the open interval (a, b)
3. $f(a) = f(b)$

Then, there is a number c in (a, b) such that $f'(c) = 0$.

ROLLE'S THEOREM

The figures show the graphs of four such functions.



Let $f(x)=x^3-2x^2+x-5$. Find the numbers c in the Rolle's theorem?

MEAN VALUE THEOREM

Equations 1 and 2

Let f be a function that fulfills two hypotheses:

1. f is continuous on the closed interval $[a, b]$.
2. f is differentiable on the open interval (a, b).

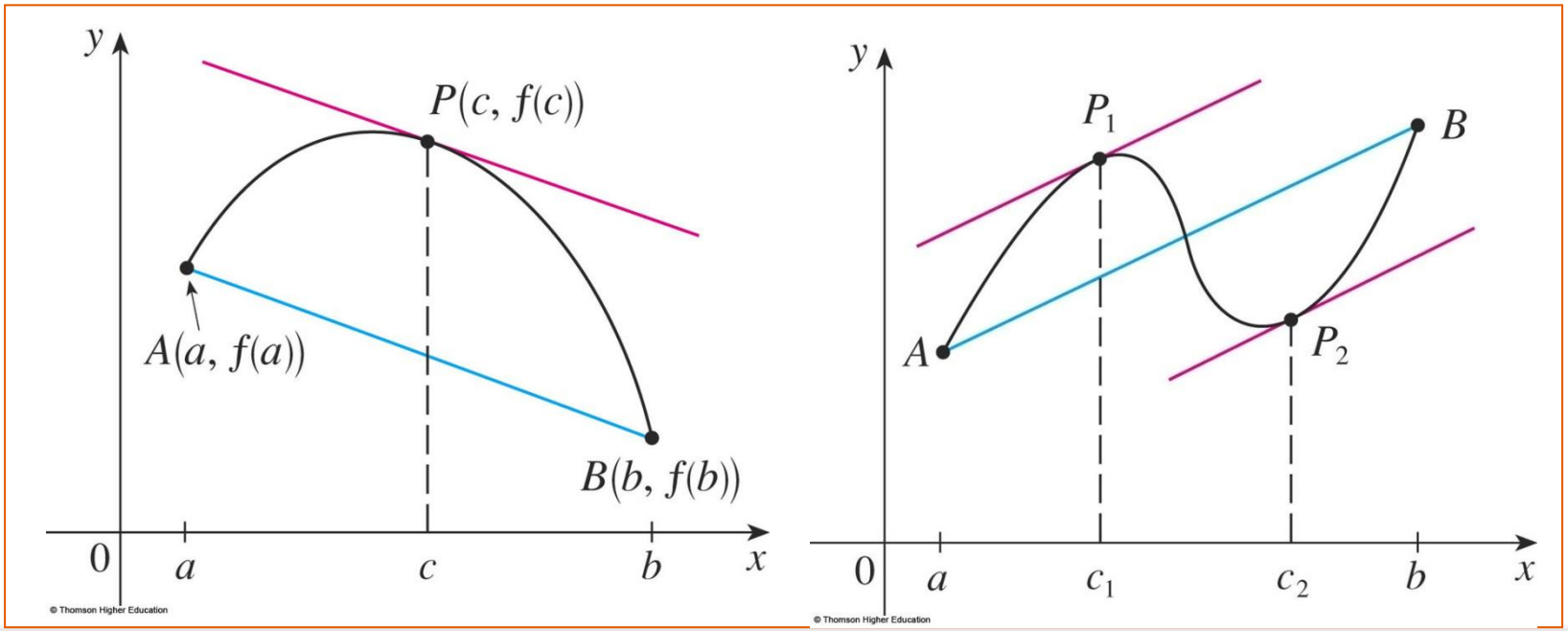
Then, there is a number c in (a, b) such that $f'(c) = \dfrac{f(b) - f(a)}{b - a}$

or, equivalently, $f(b) - f(a) = f'(c)(b - a)$

MEAN VALUE THEOREM

$f'(c)$ is the slope of the tangent line at $(c, f(c))$.

– There is at least one point $P(c, f(c))$ on the graph where the slope of the tangent line is the same as the slope of the secant line AB.



Example

- Suppose that $f(0) = -3$ and $f'(x) \leq 5$ for all values of x.
  - How large can $f(2)$ possibly be?

$\Rightarrow$ We are given that f is differentiable - and therefore continuous - everywhere.

$\Rightarrow$ In particular, we can apply the Mean Value Theorem on the interval $[0, 2]$.

- There exists a number c such that

$$f(2) - f(0) = f'(c)(2 - 0)$$

- So, $f(2) = -3 + 2f'(c)$

Example

– We are given that $f'(x) \leq 5 \ \forall x$

$\Rightarrow f'(c) \leq 5.$

$\Rightarrow 2f'(c) \leq 10.$

$\Rightarrow f(2) = -3 + 2f'(c) \leq -3 + 10 = 7$

– The largest possible value for $f(2)$ is 7.

MEAN VALUE THEOREM

Theorem

If $f'(x) = 0$ for all x in an interval (a, b), then f is constant on (a, b).

MEAN VALUE THEOREM

Corollary

If $f'(x) = g'(x)$ *for all x in an interval (a, b), then f – g is constant on (a, b).*

That is, $f(x) = g(x) + c$ *where c is a constant.*

APPLICATIONS OF DIFFERENTIATION

4.5 Derivatives and the Shapes of Graphs

In this section, we will learn:

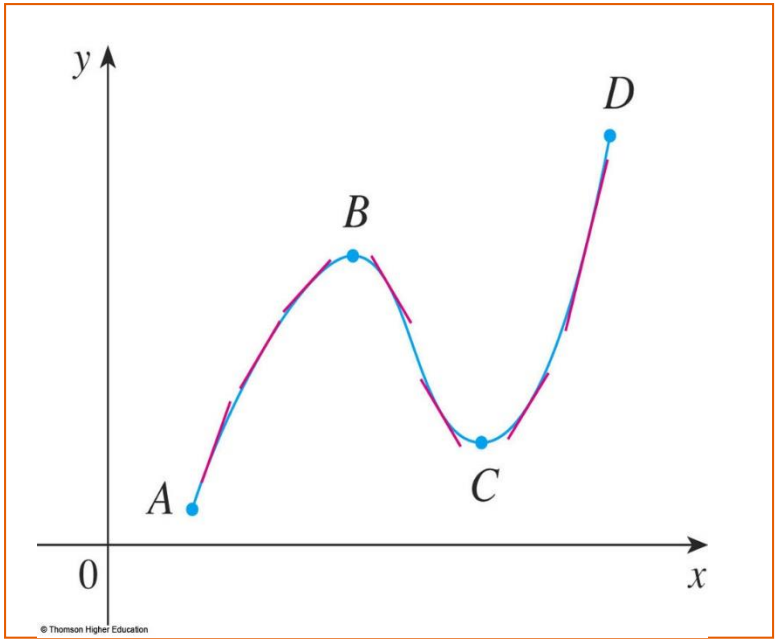
How the derivative of a function gives us the direction

in which the curve proceeds at each point.

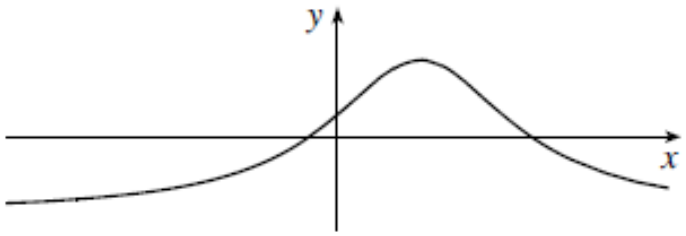
INCREASING/DECREASING TEST (I/D TEST)

a. If $f'(x) > 0$ on an interval, then f is increasing on that interval.

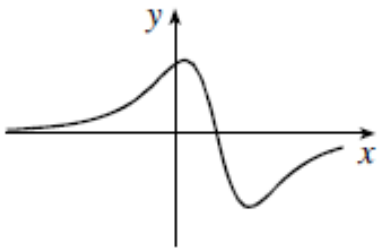
b. If $f'(x) < 0$ on an interval, then f is decreasing on that interval.



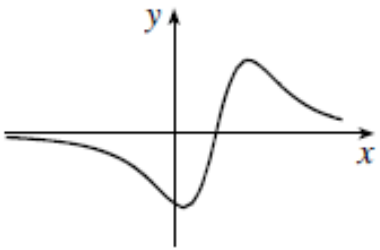
- **Drill Question:** The graph of f is shown below. Which of the following could be the graph of f'?



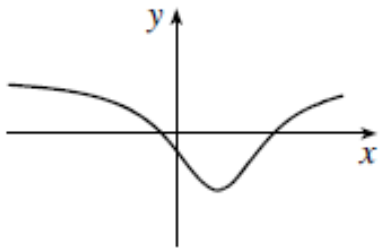
(A)



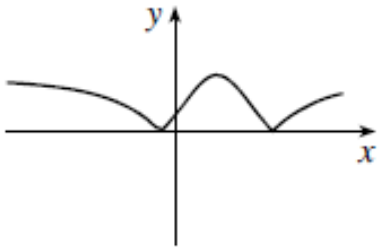
(B)



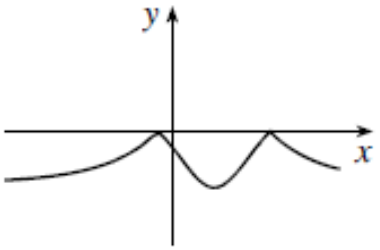
(C)



(D)



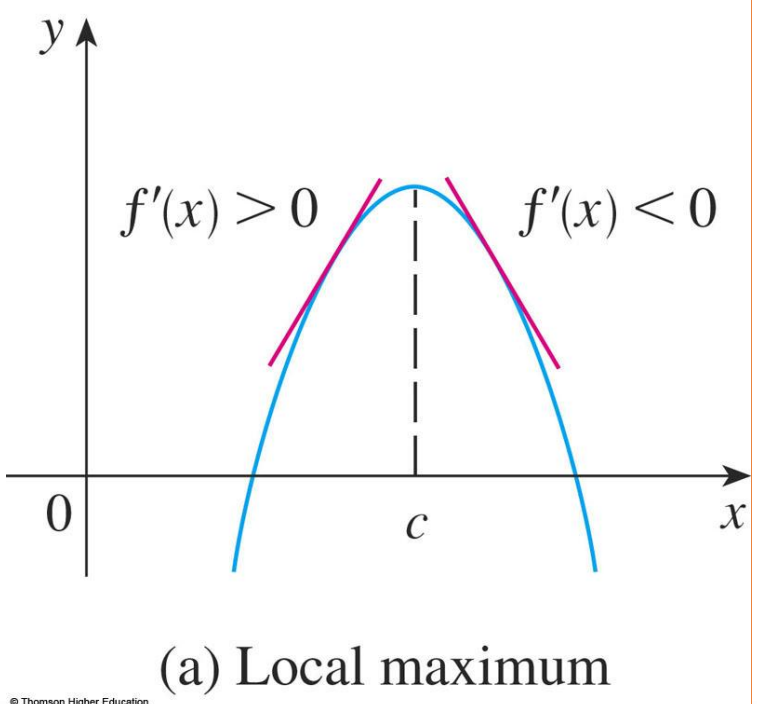
(E)



FIRST DERIVATIVE TEST

Suppose that c is a critical number of a continuous function f.

a. If f' changes from positive to negative at c, then f has a local maximum at c.

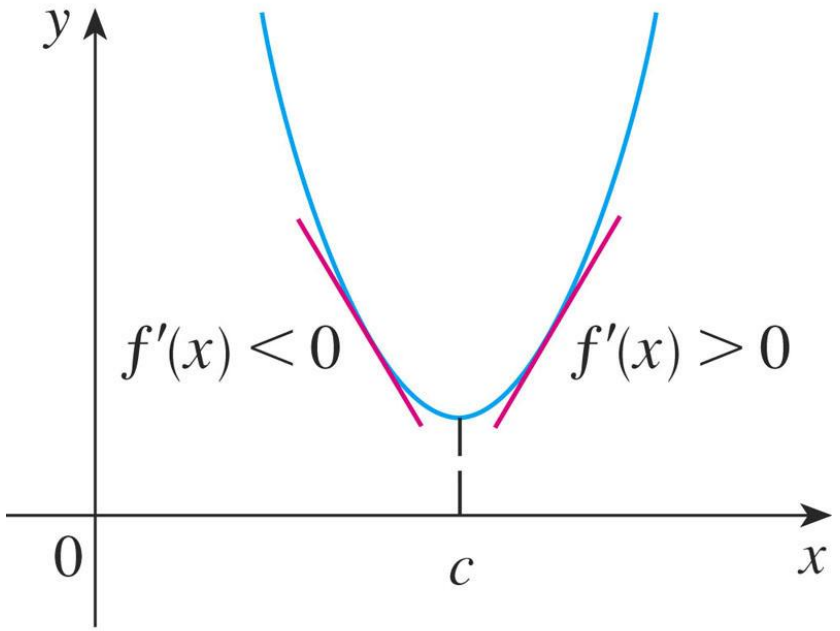


(a) Local maximum

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MEAN VALUE THEOREM

b. If f' changes from negative to positive at c, then f has a local minimum at c.

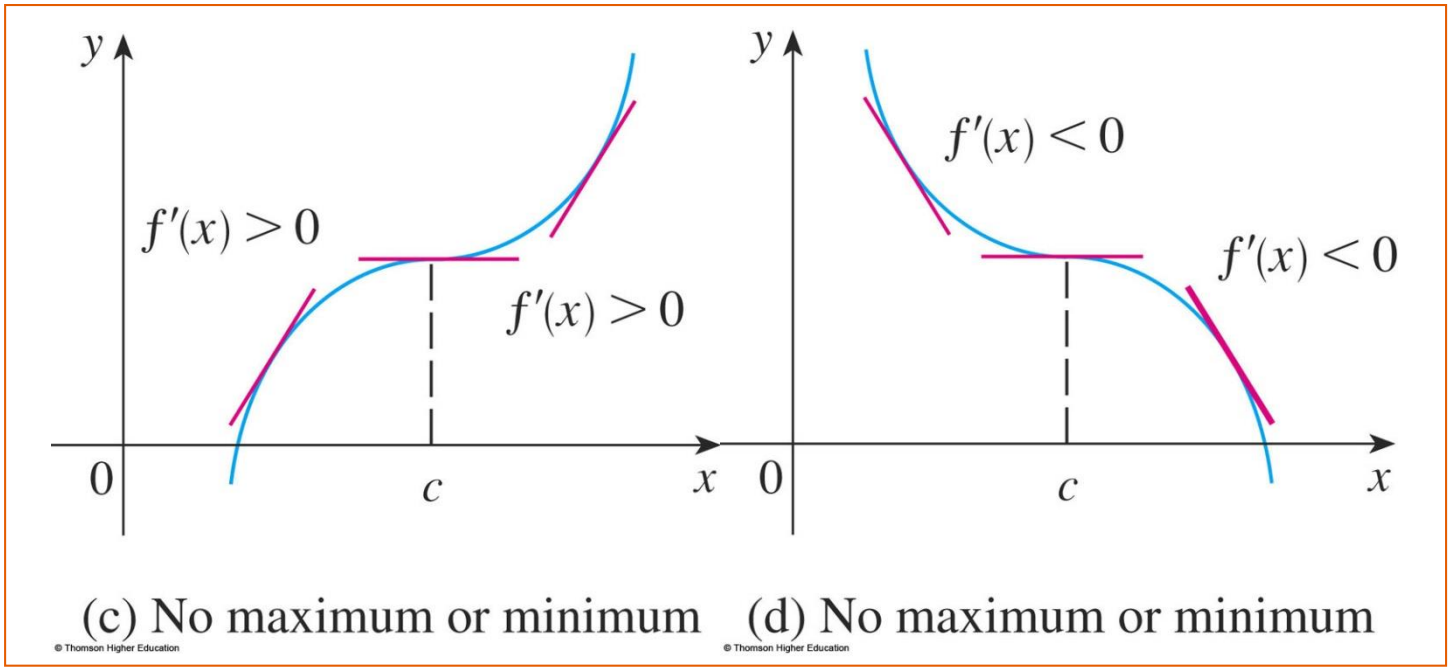


(b) Local minimum

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MEAN VALUE THEOREM

c. If f' does not change sign at c
then f has no local maximum or minimum at c.

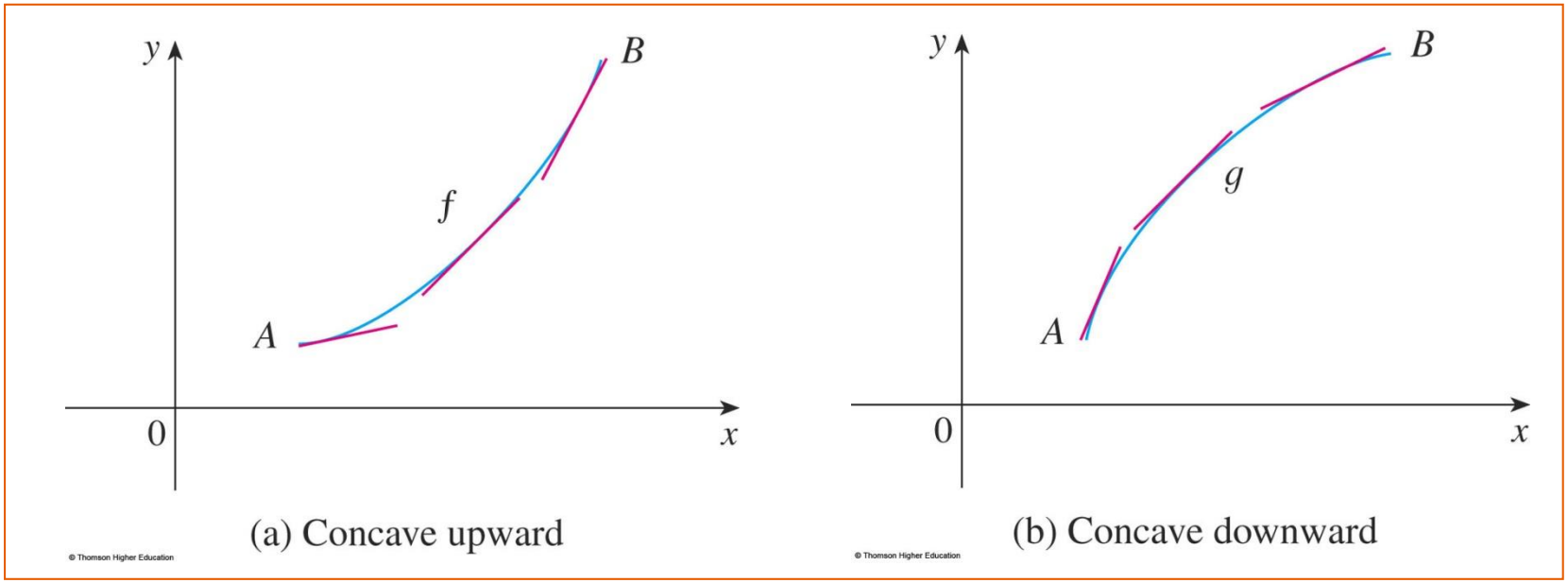


(c) No maximum or minimum (d) No maximum or minimum

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CONCAVE UPWARD/DOWNDWARD

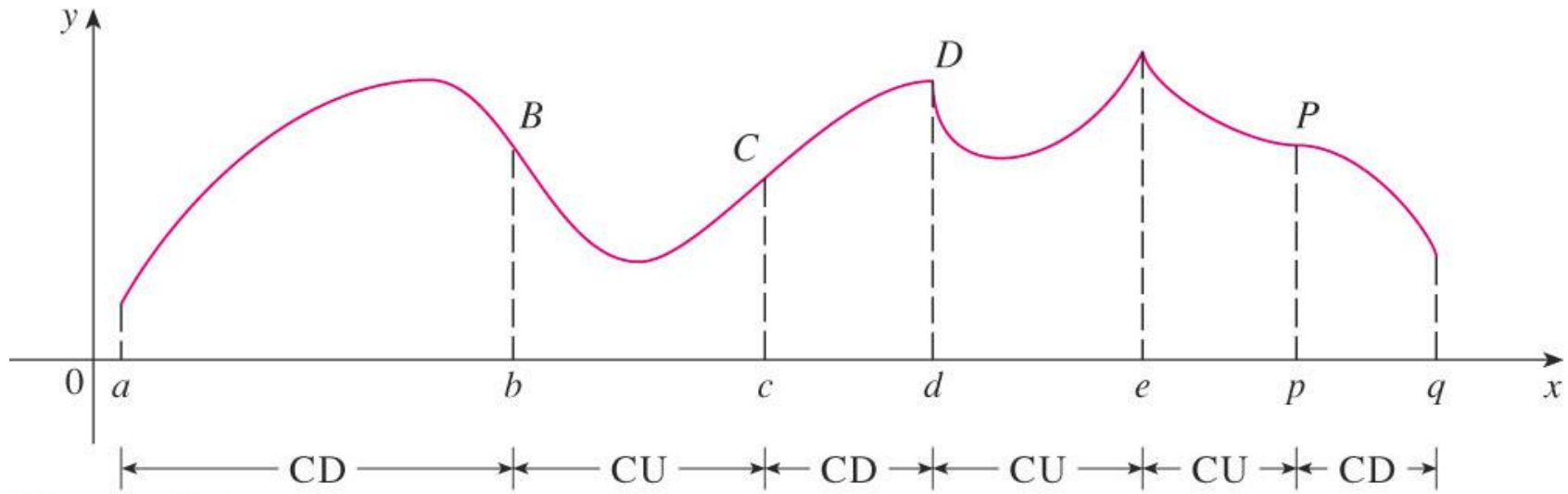
- The curve lies above the tangents and f is called **concave upward** (lõm lên) on (a, b).
- The curve lies below the tangents and g is called **concave downward** (lõm xuống) on (a, b).



CONCAVITY TEST

a. If $f''(x) > 0$ for all x in I, then the graph of f is concave upward on I.

b. If $f''(x) < 0$ for all x in I, then the graph of f is concave downward on I.



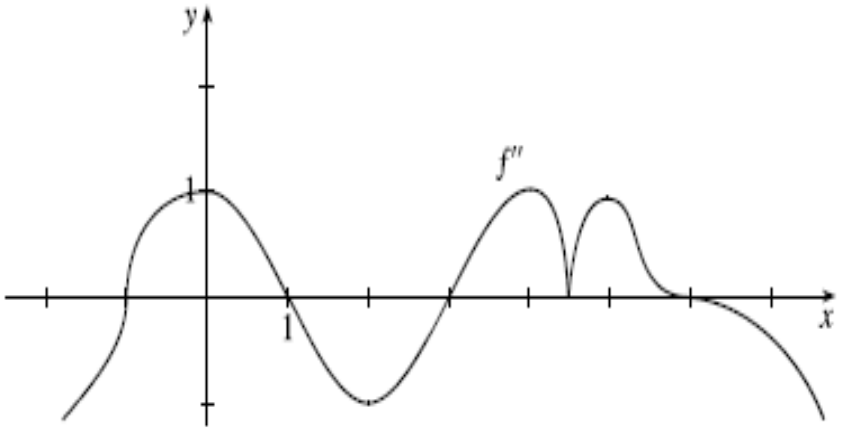
INFLECTION POINT—DEFINITION

A point P on a curve $y = f(x)$ is called an **inflection point**

if f is continuous there and the curve changes
from concave upward to concave downward

(or from concave downward to concave upward at P).

- Given a graph of f'' as below, have the students indicate the points of inflection of f, and explain their reasoning.

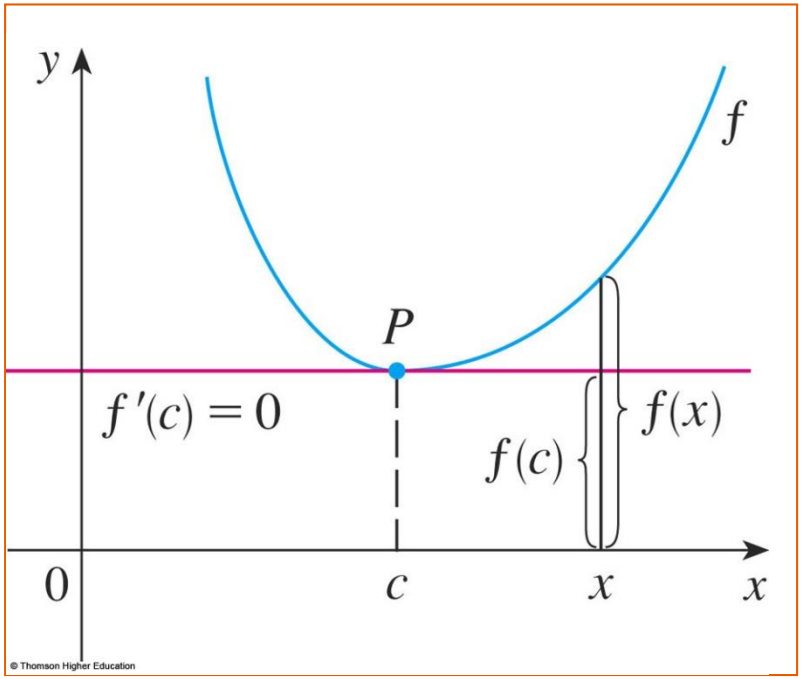


SECOND DERIVATIVE TEST

Suppose f'' is continuous near c.

a. If $f'(c) = 0$ and $\boldsymbol{f''(c) > 0}$, then f has a **local minimum** at c.

b. If $f'(c) = 0$ and $\boldsymbol{f''(c) < 0}$, then f has a **local maximum** at c.



Choose the correct one.	
A	If f has local extreme value at c then f’(c)=0.
B	If f’(c)=0 then f has local extreme value at c.
C	If f’’(3)=0 then (3,f(3)) is an inflection point of f.
D	There exists a function such that f’(x) is nonzero for all x and f(1)=f(0).
E	None of the above

APPLICATIONS OF DIFFERENTIATION

4.6 Limit at Infinity and Assymtotes

Limits at Infinity and Assymtotes

Definition

(Informal) If the values of $f(x)$ become arbitrarily close to L as x becomes sufficiently large, we say the function f has a **limit at infinity** and write

$$\lim_{x \to \infty} f(x) = L.$$

If the values of $f(x)$ becomes arbitrarily close to L for $x < 0$ as $|x|$ becomes sufficiently large, we say that the function f has a limit at negative infinity and write

$$\lim_{x \to -\infty} f(x) = L.$$

Limits at Infinity and Assymtotes

Definition

If $\lim_{x \to \infty} f(x) = L$ or $\lim_{x \to -\infty} f(x) = L$, we say the line $y = L$ is a **horizontal asymptote** of f.

Example: Find the asymptotes of the function

$$f(x) = \frac{x^3 - 1}{x^3 + x^2 - 2}$$

Example:

Compute

a. $\lim_{x\to\infty} \sin\frac{1}{x}$

b. $\lim_{x\to\infty}(\sqrt{x^2+1}-x)$

c. $\lim_{x\to\infty} \sin x$

d. $\lim_{x\to\infty}(x-x^3)$

QUIZ QUESTIONS

1) Find $\lim_{x \to \infty} \cos x$

a. 0 b. infinity c. 1 d. Does not exist

2) Find $\lim_{x \to \infty} \frac{1}{x} \cos x$

a. 0 b. infinity c. 1 d. Dose not exist

QUIZ QUESTIONS

3) If $\lim_{x\to 0} f(x) = \infty, \lim_{x\to 0} g(x) = \infty$

Then $\lim_{x\to 0}[f(x) - g(x)] = 0$

a. True b. False

4) A function can have two different horizontal asymptotes

a. True b. False

APPLICATIONS OF DIFFERENTIATION

4.7 Applied Optimization Problems

In this section, we will learn:

How to solve problems involving

maximization and minimization of factors.

1. UNDERSTAND THE PROBLEM

Read the problem carefully until it is clearly understood.

- What is the **unknown**?
- What are the **given quantities**?
- What are the given **conditions**?

2. FIND THE ABSOLUTE MAX./MIN. VALUE OF *f*

Use the methods of Sections 4.3 to find the absolute maximum or minimum value of *f*.

- In particular, if the domain of *f* is a closed interval, then the Closed Interval Method in Section 4.1 can be used.

Find two positive numbers such that the sum is 24 and the product is the largest?

Find two positive numbers such that the product is 36 and the sum is the smallest?

APPLIED OPTIMIZATION PROBLEMS

Example 1

Find the point on the line $y = 2x - 3$ that is closest to the origin.

Example 1

$(x, 2x-3)$ is in the line and the distance from it to the origin is …

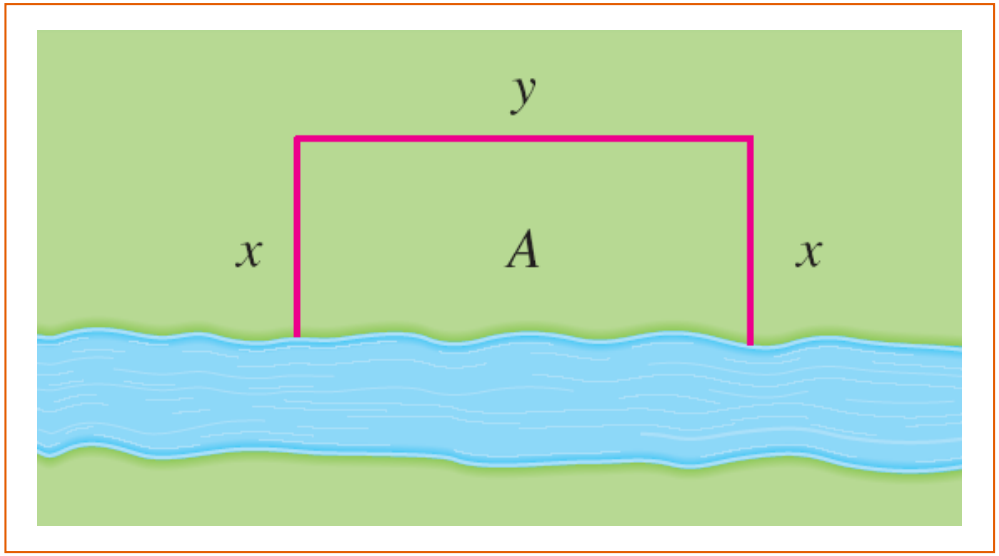
We find the minimum of…

APPLIED OPTIMIZATION PROBLEMS

Example 2

A farmer has 2400 ft of fencing and wants to fence off a rectangular field that borders a straight river. He needs no fence along the river.

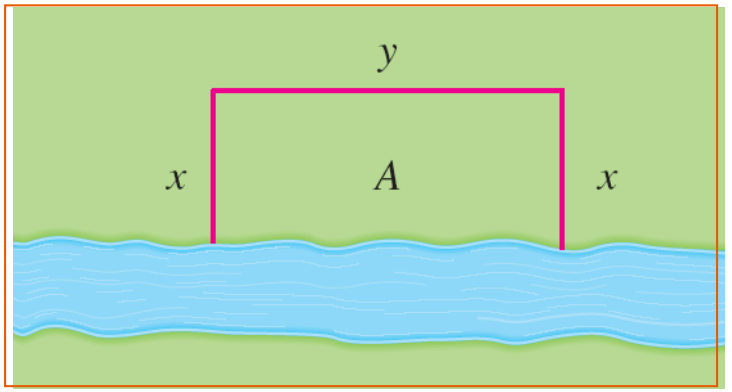
– What are the dimensions of the field that has the **largest area** ?



APPLIED OPTIMIZATION PROBLEMS

Example

This figure illustrates the general case.



We wish to maximize the area A of the rectangle.

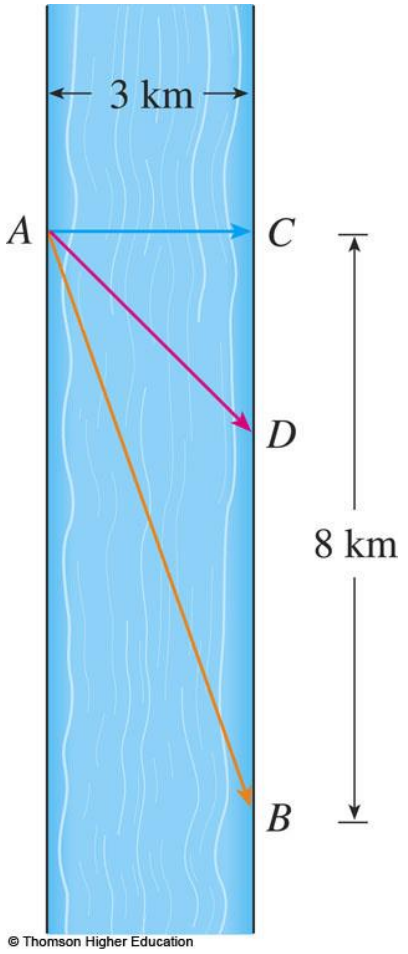
- Then, we express A in terms of x and y: $A = xy$
- $2x + y = 2400$
- So, $A(x) = 2400x - 2x^2$, $0 \leq x \leq 1200$
- …

APPLIED OPTIMIZATION PROBLEMS

Example

A man launches his boat from point A on a bank of a straight river, 3 km wide, and wants to reach point B (8 km downstream on the opposite bank) as quickly as possible.

If he can row 6 km/h and run 8 km/h, where should he land to reach B as soon as possible?



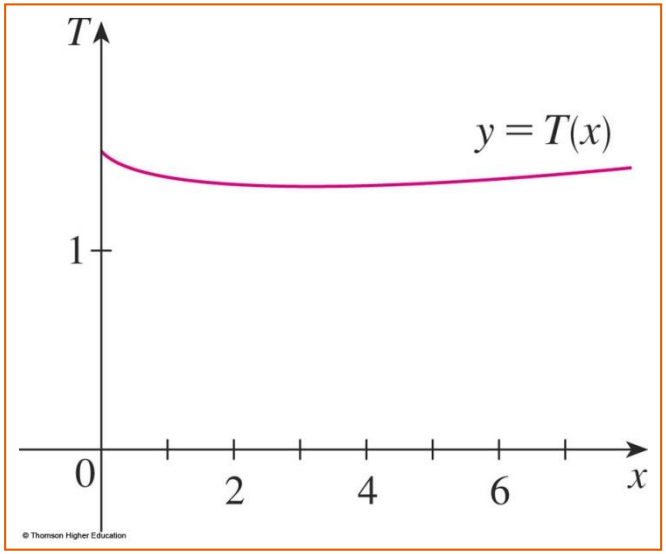
APPLIED OPTIMIZATION PROBLEMS

Example

We use the equation: $\text{time} = \dfrac{\text{distance}}{\text{rate}}$

- Then, the rowing time is: $\sqrt{x^2+9}\,/\,6$
- The running time is: $(8 - x)/8$
- So, the total time T as a function of x is:

$$T(x) = \frac{\sqrt{x^2+9}}{6} + \frac{8-x}{8}$$



A rectangular storage container with an open top is to have a volume of 15 m^3 The length of its base is twice the width.

Material for the base costs \$10 per square meter. Material for the sides costs \$6 per square meter. Find the **cost** of materials for the **cheapest** such container.

APPLICATIONS OF DIFFERENTIATION

4.9 Newton's Method

In this section, we will learn:

How to solve high-degree equations

using Newton's method.

NUMERICAL ROOTFINDERS

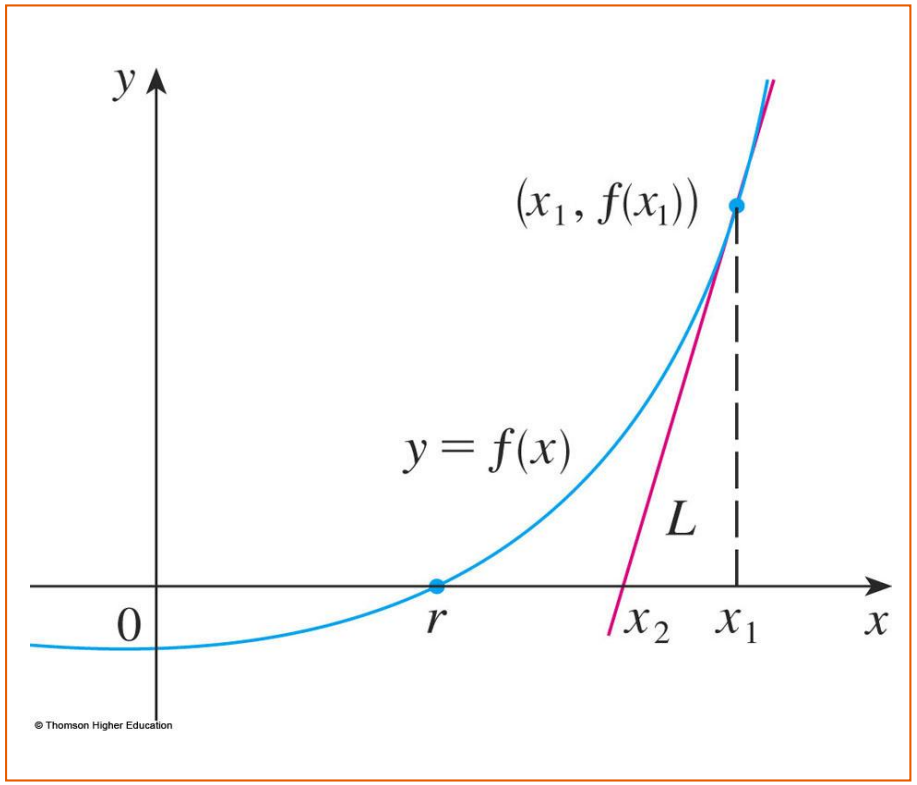
How do those numerical root finders work?

- They use a variety of methods.
- Most, though, make some use of Newton's method, also called the Newton-Raphson method.

NEWTON'S METHOD

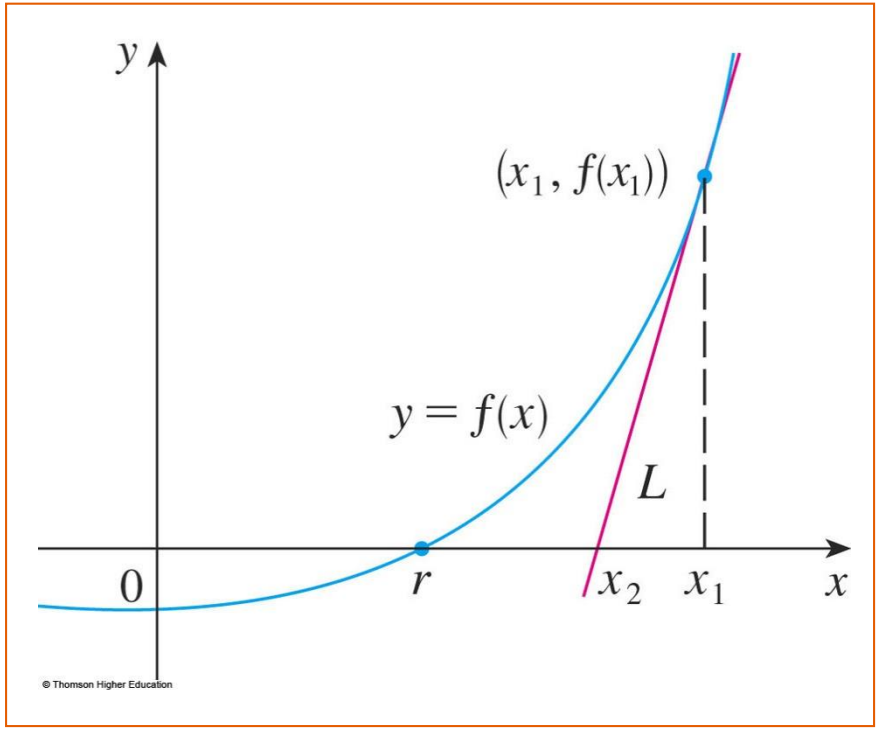
We start with a first approximation x_1, which is obtained by one of the following methods:

- Guessing
- A rough sketch of the graph of f
- A computer-generated graph of f



NEWTON'S METHOD

Consider the tangent line L to the curve $y = f(x)$ at the point $(x_1, f(x_1))$ and look at the x-intercept of L, labeled x_2.



SECOND APPROXIMATION

As the x-intercept of L is x_2, we set $y = 0$ and obtain:

$$0 - f(x_1) = f'(x_1)(x_2 - x_1)$$

If $f'(x_1) \neq 0$, we can solve this equation for x_2:

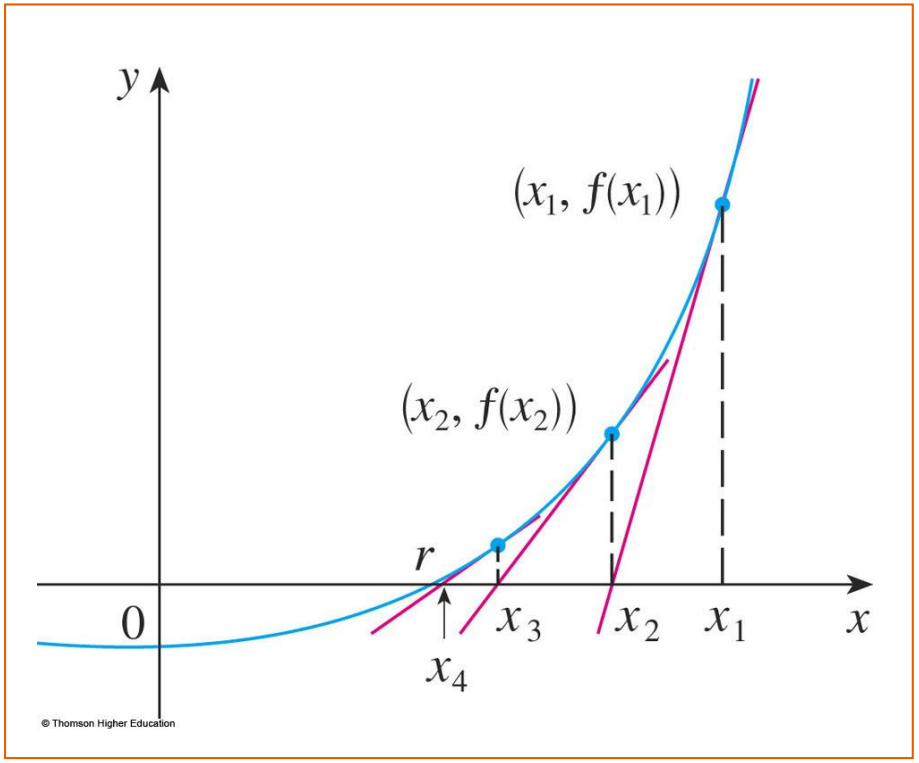
$$x_2 = x_1 - \frac{f(x_1)}{f'(x_1)}$$

We use x_2 as a second approximation to r.

SUBSEQUENT APPROXIMATION

In general, if the *n*th approximation is x_n and $f'(x_n) \neq 0$, then the next approximation is given by:

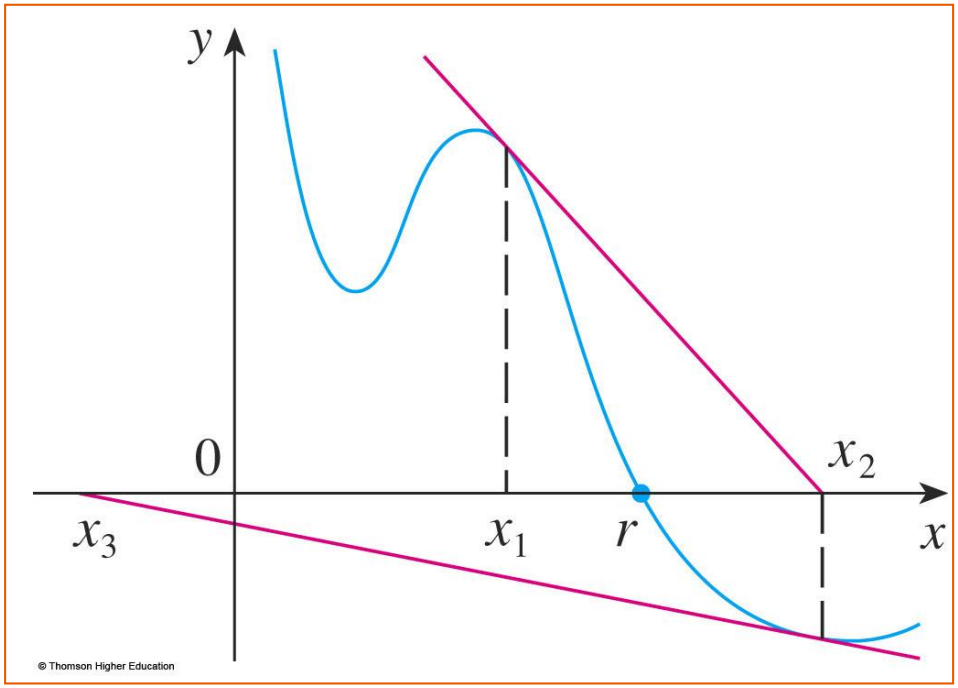
$$x_{n+1} = x_n - \frac{f(x_n)}{f'(x_n)}$$



CONVERGENCE

If the numbers x_n become closer and closer to r as n becomes large, then we say that the sequence converges to r and we write:

$$\lim_{n\to\infty} x_n = r$$



NEWTON'S METHOD

Example

Use Newton's method to find $\sqrt[6]{2}$ correct to eight decimal places.

- First, we observe that finding $\sqrt[6]{2}$ is equivalent to finding the positive root of the equation $x^6 - 2 = 0$
- So, we take $f(x) = x^6 - 2$
- Then, $f'(x) = 6x^5$

NEWTON'S METHOD

Example

So, Formula 2 (Newton's method) becomes:

$$x_{n+1} = x_n - \frac{x_n^6 - 2}{6x_n^5}$$

NEWTON'S METHOD

Example

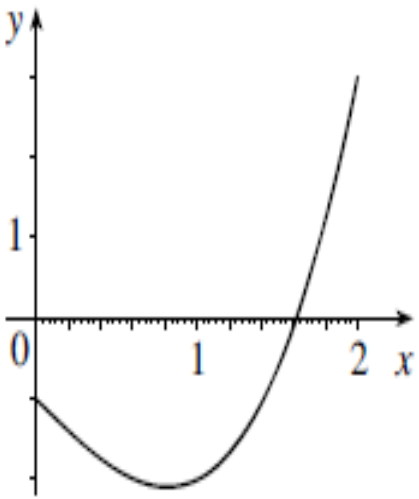
Choosing $x_1 = 1$ as the initial approximation, we obtain:

$$x_2 \approx 1.16666667$$
$$x_3 \approx 1.12644368$$
$$x_4 \approx 1.12249707$$
$$x_5 \approx 1.12246205$$
$$x_6 \approx 1.12246205$$

– As x_5 and x_6 agree to eight decimal places, we conclude that $\sqrt[6]{2} \approx 1.12246205$ to eight decimal places.

Drill Question: Use Newton's method to find x_4, the fourth approximation to the positive root of the equation $x^3 - 2x - 1 = 0$. Choose your own x_1.

Answer: The graph of $x^3 - 2x - 1$ gives 1.6 as a reasonable choice for x_1. $x_1 = 1.6 \Rightarrow x_2 \approx 1.6183099$, $x_3 \approx 1.6180341$, $x_4 \approx x_5 \approx 1.6180340$.



APPLICATIONS OF DIFFERENTIATION

4.10 Antiderivatives

In this section, we will learn about:

Antiderivatives and how they are useful

in solving certain scientific problems.

Definition

A function F is called an **antiderivative** of f on an interval I if $F'(x) = f(x)$ for all x in I.

ANTIDERIVATIVES

Theorem

If F is an antiderivative of f on an interval I, the most general antiderivative of f on I is

$$F(x) + C$$

where C is an arbitrary constant.

Let f(x)=tanx, and F(x) is an antiderivative of f(x). Evaluate f and tell whether F is increasing or decreasing at x= - 3 (rad)

Answer:

ANTIDERIVATIVE FORMULA

Here, we list some particular antiderivatives.

Function	Particular antiderivative	Function	Particular antiderivative
$cf(x)$	$cF(x)$	$\sin x$	$-\cos x$
$f(x) + g(x)$	$F(x) + G(x)$	$\sec^2 x$	$\tan x$
$x^n \quad (n \neq -1)$	$\dfrac{x^{n+1}}{n+1}$	$\sec x \tan x$	$\sec x$
$1/x$	$\ln \lvert x \rvert$	$\dfrac{1}{\sqrt{1-x^2}}$	$\sin^{-1} x$
e^x	e^x	$\dfrac{1}{1+x^2}$	$\tan^{-1} x$
$\cos x$	$\sin x$		

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RECTILINEAR MOTION

Example

A particle moves in a straight line and has acceleration given by $a(t) = 6t + 4$.

Its initial velocity is $v(0) = -6$ cm/s and its initial displacement is $s(0) = 9$ cm.

– Find its position function $s(t)$.

A particle moves along the x-axis so that its velocity at time t is given by 3 sin 2t.
Assuming it starts at the origin, where is it at $t = \pi$ seconds?

a. 0 b. 3/2 c. ½ d. - 1/2

Let f(x)=4-3x for all x>=2. Select the correct one.

a. 2 is the local minimum value.
b. 2 is the local maximum value.
c. -2 is the local minimum value.
d. -2 is the maximum local value.
e. 2 is the absolute minimum value.
f. -2 is the absolute maximum value.
g. None of the above.

A piece of wire (dây kim loại) 10 m long is cut into two pieces. One piece is bent into a square and the other is bent into an equilateral triangle (tam giác đều).
How should the wire be cut for the square so that the total area (of the square and the triangle) enclosed is a minimum? Round the result to the nearest hundredth.

Answer: If x is the length of the square then

the side of the triangle is 1/3 (10-4x)

We find the minimum of x^2+…